

# Microsoft Azure

## Azure Data Factory - Control Flow Activity Project

---

Aishwarya Mohan

# Introduction

As we have done earlier (Resource Group, Storage Account, Data Factory creation, Source and Destination containers, and the Copy Activity foundation), we now continue with Control Flow activities inside Azure Data Factory Studio.

This project covers the following Control Flow activities and triggers:

- Copy Data activity with column mapping
- Delete activity with logging settings
- Triggers: Manual, Schedule, Tumbling Window, and Storage Events
- Set Variable – area calculation (fixed values and parameterised runtime input)
- Append Variable
- Get Metadata
- Execute Pipeline

All activities were configured and executed on the Microsoft Azure platform.  
The screenshots below show the exact configuration and successful runs.

## Step 1: Copy Data Activity with Mapping

Open pipeline1. Add Copy data activity (Source to Destination).

Source dataset: SD\_01 (Pokemon.csv). Sink dataset: DS\_01.

Mapping tab – map columns: No, Name, Type1, Type2, Height, Weight, Legendary.

The screenshot displays the Microsoft Azure Data Factory console. The left sidebar shows the 'Factory Resources' tree with 'Pipelines' and 'Datasets' sections. 'pipeline1' is selected under Pipelines, and 'SD\_01' is selected under Datasets. The main workspace shows the 'Activities' list on the left and the 'Copy data' activity configuration in the center. The 'Copy data' activity is configured with the following settings:

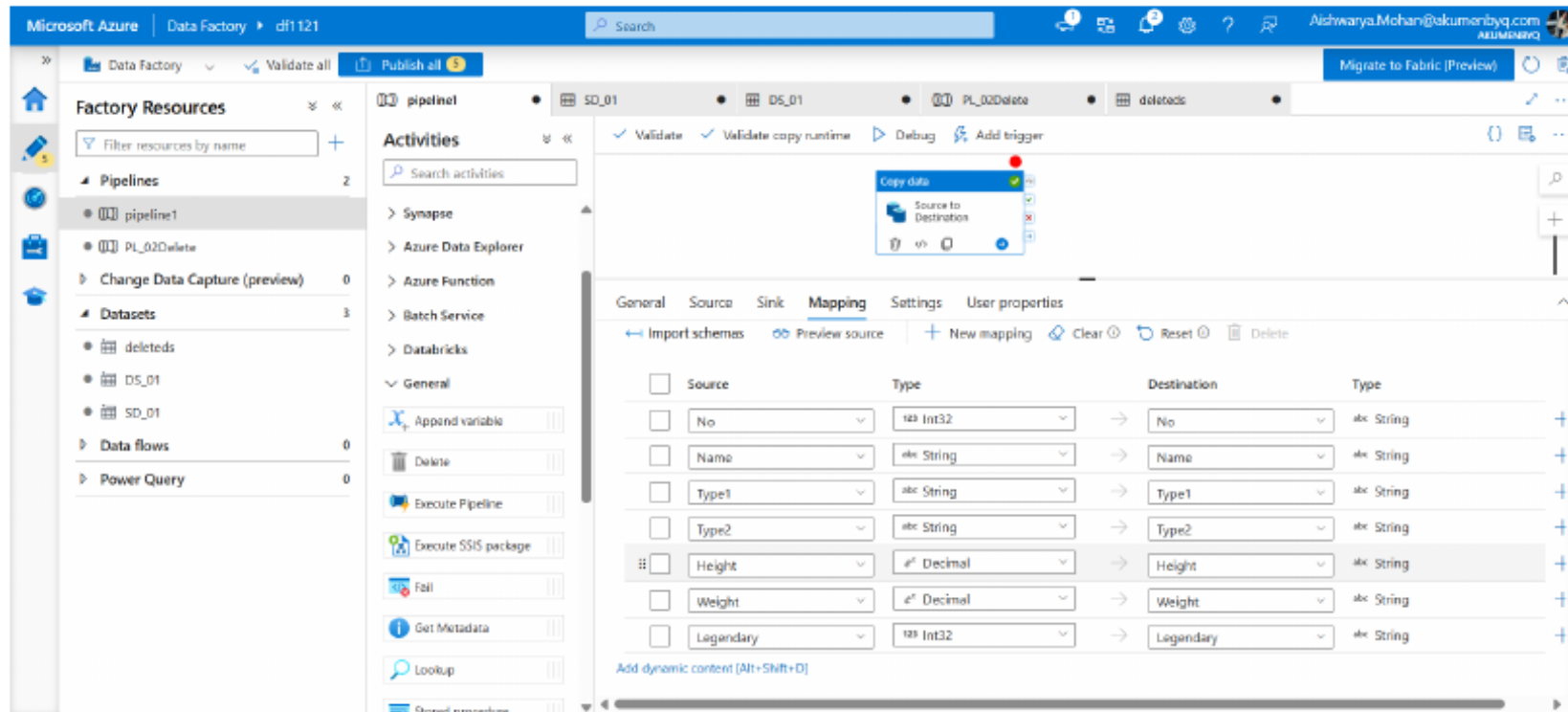
- Source dataset:** SD\_01
- File path type:** File path in dataset
- Start time (UTC):** (Empty field)
- End time (UTC):** (Empty field)
- Filter by last modified:** (Empty field)
- Recursively:** ☒
- Enable partitions discovery:** ☐

The 'Mapping' tab is currently selected, showing the 'Source to Destination' mapping interface. The 'General' tab is also visible, showing the 'Copy data' activity name and the 'Source to Destination' mapping.

## Step 1 (continued): Column Mapping Configuration

Import schemas and map source columns to destination columns.

Type conversions are applied as shown (Int32 → String, Decimal → String, etc.).



The screenshot displays the Microsoft Azure Data Factory interface. The top navigation bar shows 'Microsoft Azure | Data Factory | df1121'. The left sidebar contains 'Factory Resources' with sections for Pipelines (2 items), Datasets (3 items), Data flows (0 items), and Power Query (0 items). The central pane shows 'Activities' with a search bar and a list of activity types: Synapse, Azure Data Explorer, Azure Function, Batch Service, and Databricks. The main workspace shows a 'Copy data' activity with a 'Mapping' tab selected. The mapping table shows source columns being mapped to destination columns with type conversions.

| Source    | Type    | Destination | Type   |
|-----------|---------|-------------|--------|
| No        | Int32   | No          | String |
| Name      | String  | Name        | String |
| Type1     | String  | Type1       | String |
| Type2     | String  | Type2       | String |
| Height    | Decimal | Height      | String |
| Weight    | Decimal | Weight      | String |
| Legendary | Int32   | Legendary   | String |

## Step 2: Successful Copy Data Pipeline Run

Debug or Trigger the pipeline. Pipeline status: Succeeded.

Activity 'Source to Destination' completed successfully. File appears in destination container.

The screenshot displays the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane shows a list of pipelines, including 'pipeline1' and 'PL\_02Delete'. The main workspace shows the 'pipeline1' canvas with a 'Copy data' activity. The 'Activities' pane on the left lists various activities like Synapse, Azure Data Explorer, and Batch Service. The right pane shows the 'Output' tab for the 'Copy data' activity, indicating a successful pipeline run with the status 'Succeeded'.

**Factory Resources**

- Pipelines: 2
  - pipeline1
  - PL\_02Delete
- Change Data Capture (preview): 0
- Datasets: 3
  - deleted
  - DS\_01
  - SD\_01
- Data flows: 0
- Power Query: 0

**Activities**

- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- General
  - Append variable
  - Delete
  - Execute Pipeline
  - Execute SSIS package
  - Fail
  - Get Metadata
  - Lookup
  - Stored procedure

**Copy data**

Source to Destination

**Pipeline run** 6d8b7891-789e-4e87-a387-ef1e2 1b1178f **Pipeline status** Succeeded

Showing 1 - 1 of 1 items

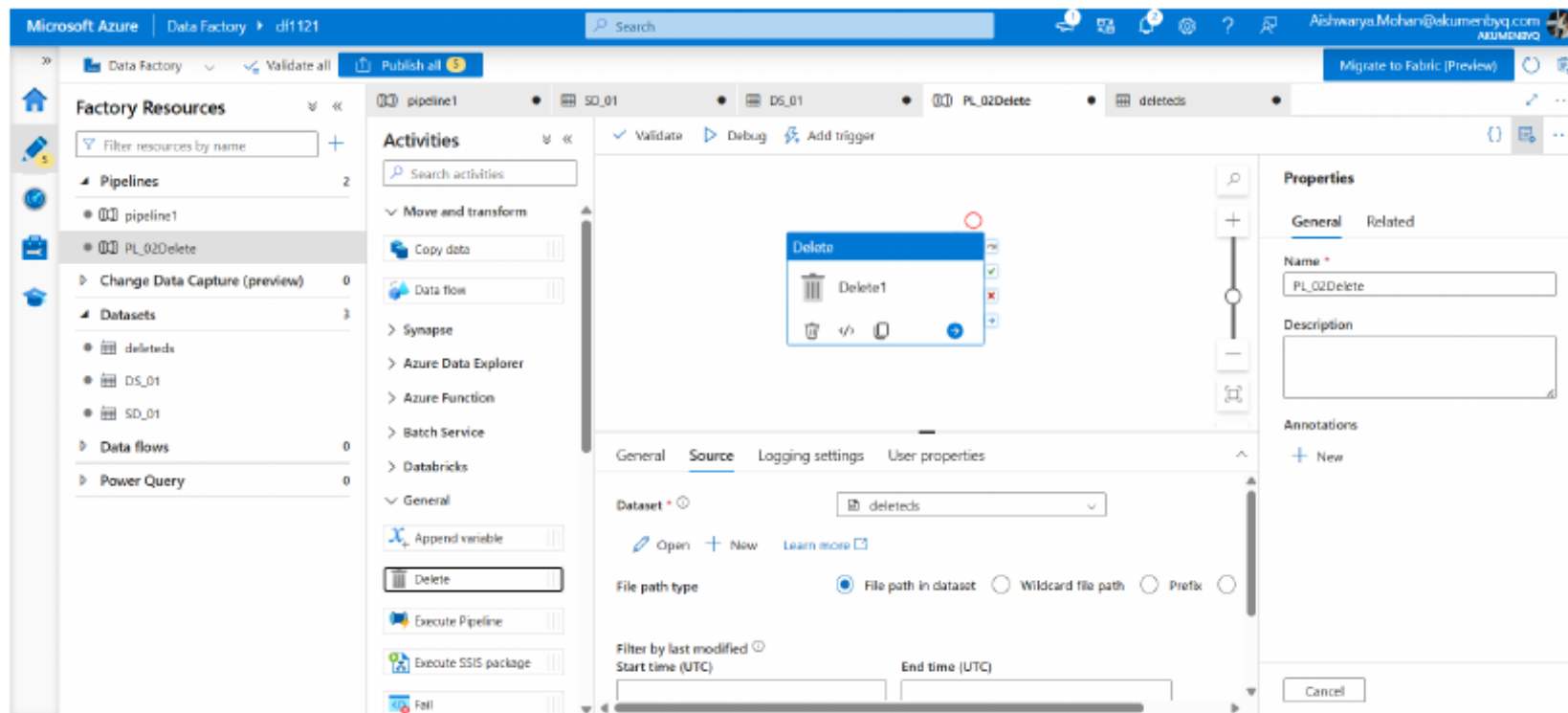
| Activity name         | Activity st... | Activit... | Run start              | Duration | Integration runtime                     |
|-----------------------|----------------|------------|------------------------|----------|-----------------------------------------|
| Source to Destination | Succeeded      | Copy data  | 8/14/2026, 12:53:30 PM | 17s      | AutoResolveIntegrationRuntime (East US) |

## Step 3: Delete Activity

Create pipeline PL\_02Delete (or add Delete activity).

Configure Delete activity with dataset 'deletedes' pointing to ABSTRACT.docx.

File path type: File path in dataset.



## Step 3 (continued): Delete Activity Successful Run

Pipeline status: Succeeded. The target file is deleted from the container.

The screenshot displays the Microsoft Azure Data Factory interface. The top navigation bar shows 'Microsoft Azure | Data Factory | df1121'. The left sidebar lists 'Factory Resources' including Pipelines (2), Datasets (3), Data flows (0), and Power Query (0). The 'Pipelines' section is expanded, showing 'pipeline1' and 'PL\_Q2Delete'. The 'Activities' section is also expanded, showing 'Delete' and 'Delete1'. The main canvas displays the 'Delete' activity with a green checkmark, indicating success. The 'Output' tab is selected, showing the pipeline status as 'Succeeded' with a green checkmark. The pipeline ID is '79095caf-a679-4701-85b6-d0acc55f2d4c'. The activity name is 'Delete1', and the status is 'Succeeded'. The run start time is '8/14/2026, 12:57:21 PM'. The 'Properties' panel on the right shows the 'General' tab with the name 'PL\_Q2Delete' and a description field. The 'Annotations' section has a '+ New' button. A 'Cancel' button is located at the bottom right of the output panel.

Microsoft Azure | Data Factory | df1121

Search

Validate all Publish all

Migrate to Fabric (Preview)

Factory Resources

Filter resources by name

Pipelines 2

- pipeline1
- PL\_Q2Delete

Change Data Capture (preview) 0

Datasets 3

- deleted1
- DS\_01
- SD\_01

Data flows 0

Power Query 0

Activities

Search activities

Move and transform

- Copy data
- Data flow

Synapse

Azure Data Explorer

Azure Function

Batch Service

Databricks

General

- Append variable
- Delete
- Execute Pipeline
- Execute SSIS package
- Fail

Validate Debug Add trigger

Delete

Delete1

Parameters Variables Settings Output

Pipeline 79095caf-a679-4701-85b6-d0acc55f2d4c

run ID 5b6-d0acc55f2d4c

Pipeline status Succeeded

View debug run consumption

All status

Monitor in Azure Metrics Export to CSV

Showing 1 - 1 of 1 items

| Activity name | Activity st... | Activit... | Run start              |
|---------------|----------------|------------|------------------------|
| Delete1       | Succeeded      | Delete     | 8/14/2026, 12:57:21 PM |

Properties

General Related

Name PL\_Q2Delete

Description

Annotations

+ New

Cancel

## Step 4: Delete Activity – Logging Settings

Enable logging on the Delete activity.

Select logging account linked service and set Folder path (e.g. source).

This keeps a log of deleted files for audit purposes.

The screenshot displays the Microsoft Azure Data Factory portal. The top navigation bar shows 'Data Factory' and 'df1121'. The left sidebar lists 'Factory Resources' including Pipelines (pipeline1, PL\_02Delete), Datasets (deleted1, DS\_01, SD\_01), Data flows, and Power Query. The main canvas shows the 'Delete' activity 'Delete1' in the 'PL\_02Delete' pipeline. The 'Logging settings' tab is active, showing 'Enable logging' checked, 'Logging account linked service' set to 'LS\_SD', and 'Folder path' set to 'source'. A 'Delete' dialog box is visible over the activity, showing a trash icon and the name 'Delete1'.

## Step 5: Manual Trigger (Trigger now / Debug)

Use Debug or 'Trigger now' to execute the pipeline manually.

View successful pipeline run and activity output in the monitoring pane.

The screenshot displays the Microsoft Azure Data Factory console. The top navigation bar shows 'Data Factory' and 'df1121'. The left sidebar, 'Factory Resources', lists 'Pipelines' (2 items) and 'Datasets' (3 items). The 'Pipelines' section is expanded, showing 'pipeline1', 'PL\_02Delete', and 'Change Data Capture (preview)'. The 'Activities' pane on the right lists various activities like 'Synapse', 'Azure Data Explorer', 'Azure Function', 'Batch Service', 'Databricks', 'Append variable', 'Delete', 'Execute Pipeline', 'Execute SSIS package', 'Fail', 'Get Metadata', 'Lookup', and 'Stored procedure'. The main canvas shows a pipeline diagram with activities 'SD\_01', 'DS\_01', 'PL\_02Delete', and 'deletedata'. A context menu is open over the 'Copy data' activity, with the 'Trigger now' option selected. The bottom pane shows the 'General' tab for the 'Source to Destination' activity, with fields for 'Name', 'Description', 'Activity state' (set to 'Activated'), 'Timeout' (0.12:00:00), 'Retry' (0), and 'Retry interval (sec)' (30).

## Step 6: Schedule Trigger

Create a new Schedule trigger (trigger1).

Set Start date/time, Time zone (e.g. Chennai, Kolkata, Mumbai, New Delhi), and Recurrence (e.g. every 15 minutes).

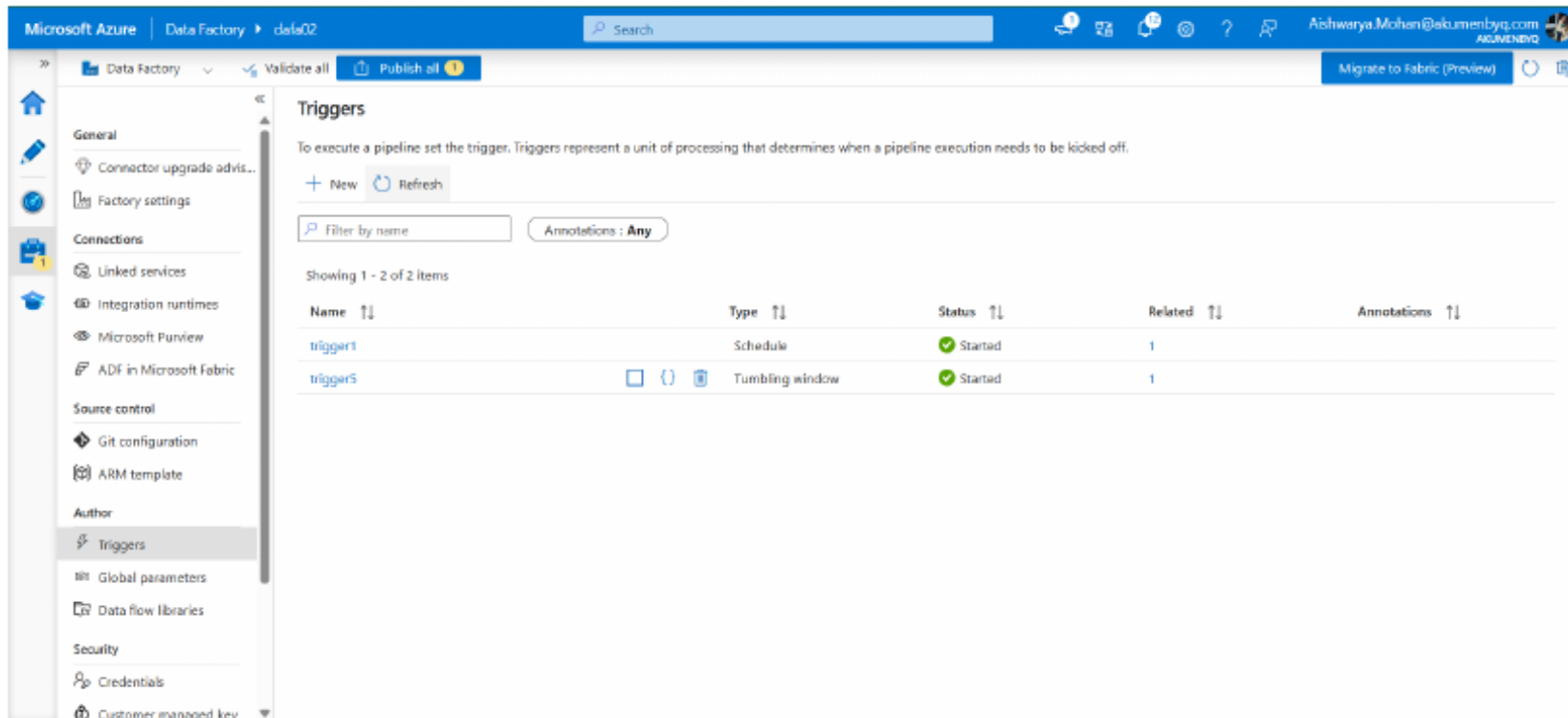
Publish all changes and start the trigger.

The screenshot displays the Microsoft Azure Data Factory portal interface. On the left, the 'Factory Resources' pane shows a hierarchy: Pipelines (1), Datasets (2), Data flows (0), and Power Query (0). The 'Activities' pane lists various services like Move and transform, Synapse, Azure Data Explorer, etc. The main workspace shows a pipeline named 'pipeline1' with two datasets, 'DelimitedTest1' and 'DelimitedTest2'. A 'Copy data' activity is visible in the workspace. On the right, the 'New trigger' dialog box is open. It contains the following fields: Name (trigger1), Description (empty), Type (Schedule), Start date (8/17/2026, 9:24:33 PM), Time zone (Chennai, Kolkata, Mumbai, New Delhi (UTC+5:30)), Recurrence (Every 15 Minute(s)), and a checkbox for 'Specify an end date' (unchecked). There is also a section for 'Annotations' with a '+ New' button, and a 'Start trigger' checkbox (checked) with a note 'Start trigger on creation'. At the bottom of the dialog are 'OK' and 'Cancel' buttons.

## Step 7: Tumbling Window Trigger

Create a Tumbling window trigger.

Configure the window size and start it. Status appears as 'Started' in the Triggers list.



The screenshot shows the Microsoft Azure Data Factory interface. The top navigation bar includes the Microsoft Azure logo, the Data Factory name 'dels02', a search bar, and user information 'Aishwarya.Mohan@ekumenbyg.com'. The left sidebar contains a navigation menu with categories like General, Connections, Source control, Author, and Security. The 'Triggers' option under the Author category is selected. The main content area is titled 'Triggers' and includes a description: 'To execute a pipeline set the trigger. Triggers represent a unit of processing that determines when a pipeline execution needs to be kicked off.' Below the description are buttons for '+ New' and 'Refresh'. A filter bar shows 'Filter by name' and 'Annotations: Any'. The table below shows 1 - 2 of 2 items:

| Name     | Type            | Status  | Related | Annotations |
|----------|-----------------|---------|---------|-------------|
| trigger1 | Schedule        | Started | 1       |             |
| trigger5 | Tumbling window | Started | 1       |             |

## Step 8: Storage Event Trigger

Create a Storage events trigger (Blob created).

Select Azure subscription, Storage account, and Container (source).

Publish and start. Upload a file to the source container to fire the trigger.

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane shows a hierarchy: Data Factory > df1121 > Pipelines > pipeline1 > Datasets > DelimitedText1. The 'Activities' pane shows a list of activities including 'Move and transform', 'Synapse', 'Azure Data Explorer', 'Azure Function', 'Batch Service', 'Databricks', 'General', 'HDInsight', 'Iteration & conditionals', 'Machine Learning', and 'Power Query'. The 'Output' pane shows the 'Pipeline run ID' as 719c5db9-d757-4f4d-8c31-7228a785ae4f and the 'Activity name' as 'Copy data1', which has a status of 'Succeeded'. The 'Edit trigger' pane on the right is configured with the following details:

- Name:** trigger1
- Description:** (empty)
- Type:** BlobEventsTrigger
- Account selection method:** From Azure subscription (selected), Enter manually (unselected)
- Azure subscription:** Azure subscription 1 (eab6ae60-ef6-4e9f-8597-8bb556d42eb7)
- Storage account name:** saveacco
- Container name:** source
- Blob path begins with:** (empty)

At the bottom of the 'Edit trigger' pane are 'Continue' and 'Cancel' buttons.

## Step 9: Set Variable – Area Calculation (Fixed Values)

Create pipeline2. Add three Set variable activities: length, breadth, area.

Set length = 10, breadth = 10.

For area use the expression: `@mul(variables('length'), variables('breadth'))`

The screenshot displays the Microsoft Azure Data Factory portal interface. The top navigation bar shows 'Microsoft Azure' and 'Data Factory' with the instance name 'df1121'. A search bar and user profile 'Ashwarya.Mohan@akumenbyg.com' are also visible. Below the navigation bar, a banner promotes upgrading to Microsoft Fabric Data Factory. The main workspace is divided into three panels: 'Factory Resources', 'Activities', and 'Properties'.

**Factory Resources:** This panel on the left lists the available resources. Under 'Pipelines', 'pipeline2' is selected and highlighted.

**Activities:** The central panel shows the 'pipeline2' canvas. It contains three 'Set variable' activities arranged vertically. The top activity is for 'length', the middle for 'breadth', and the bottom for 'area'. Each activity has a blue 'Set variable' icon and a variable name in a text box. The 'area' activity's expression is `@mul(variables('length'), variables('breadth'))`. The 'Activities' list on the left includes 'Execute Pipeline', 'Execute SSIS package', 'Fail', 'Get Metadata', 'Lookup', 'Stored procedure', 'Script', 'Set variable', 'Validation', 'Web', 'WebHook', and 'Wait'.

**Properties:** The right panel shows the 'Properties' tab for the selected 'Set variable' activity. It has sub-tabs for 'General', 'Settings', and 'User properties'. The 'General' tab is active, showing the 'Name' field set to 'length' and a 'Description' field. The 'Activity state' is set to 'Activated' (radio button selected). The 'Secure output' checkbox is unchecked. A 'Cancel' button is at the bottom right.

## Step 9 (continued): Setting Fixed Values and Expression

length variable is set to the value 10.

area variable uses the multiply expression on length and breadth variables.

The screenshot displays the Microsoft Azure Data Factory portal. The left sidebar shows the 'Factory Resources' tree with 'pipeline2' selected under 'Control flow activity'. The main canvas shows the 'Activities' list with 'Set variable' selected. The 'Set variable' activity is configured with the following settings:

- Variable type:** Pipeline variable (selected)
- Name:** length
- Value:** 10

The 'Properties' pane on the right shows the 'General' tab with the following details:

- Name:** pipeline2
- Description:** (empty)
- Annotations:** + New

The 'Settings' tab is also visible, showing the 'Variable type' and 'Name' fields. The 'Value' field is set to '10'.

## Step 9 (continued): Expression Builder for Area

Pipeline expression builder: @mul(variables('length'), variables('breadth'))

This calculates area = length × breadth.

The screenshot displays the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane shows a list of pipelines, with 'pipeline2' selected. The 'Activities' pane on the right shows a sequence of activities: 'Set variable' (length), 'Set variable' (breadth), and 'Set variable' (area). The 'Settings' tab for the 'Set variable' activity is active, showing the variable name 'area' and the value field. The 'Pipeline expression builder' dialog is open, showing the expression '@mul(variables('length'),variables('breadth'))' in the text area. Below the text area, the 'Variables' tab is selected, showing a list of variables: 'area', 'breadth', and 'length'. A 'click to insert expression' button is visible next to the 'length' variable. The 'OK' and 'Cancel' buttons are at the bottom of the dialog.

Microsoft Azure | Data Factory | df1121

Explore upgrade options to Microsoft Fabric Data Factory. Get started with guided support. [Learn more](#)

Factory Resources

- Pipelines
- pipeline1
- Control flow activity
- pipeline2
- Change Data Capture (preview)
- Datasets
- Data flows
- Power Query

Activities

- Search activities
- Execute Pipeline
- Execute SSIS package
- Fail
- Get Metadata
- Lookup
- Stored procedure
- Script
- Set variable
- Validation
- Web
- WebHook
- Wait

pipeline1 | pipeline2

Validate | Debug | Add trigger

Set variable

(x) length

Set variable

(x) breadth

Set variable

(x) area

General | Settings | User properties

Variable type: Pipeline variable

Name: area

Value: @mul(variables('length'),variables('breadth'))

Pipeline expression builder

Add dynamic content below using any combination of expressions, functions and system variables.

@mul(variables('length'),variables('breadth'))

Clear contents

Parameters | System variables | Functions | Variables

Search

area

breadth

length

click to insert expression

OK | Cancel

## Step 9 (continued): Successful Run – Area = 100

All three Set variable activities succeeded.

With length=10 and breadth=10 the calculated area is 100.

The screenshot displays the Microsoft Azure Data Factory console. The left sidebar shows 'Factory Resources' with 'Pipelines' expanded, listing 'pipeline1' and 'pipeline2'. The main canvas shows the 'pipeline2' design, which includes three 'Set variable' activities: 'length', 'breadth', and 'area'. All three activities are marked with a green checkmark, indicating success. The 'Properties' pane on the right shows the 'General' tab for 'pipeline2'. The 'Output' tab at the bottom displays a table of activity results.

| Activity name | Activity status | Activity...  | Run start             |
|---------------|-----------------|--------------|-----------------------|
| area          | ✓ Succeeded     | Set variable | 8/17/2026, 2:28:02 PM |
| breadth       | ✓ Succeeded     | Set variable | 8/17/2026, 2:28:02 PM |
| length        | ✓ Succeeded     | Set variable | 8/17/2026, 2:28:02 PM |

## Step 10: Set Variable – Parameterised Area Calculation

Add pipeline parameters: length, breadth (and optionally area).

Change Set variable activities to use @pipeline().parameters.length and @pipeline().parameters.breadth. Supply values at pipeline run time.

The screenshot displays the Microsoft Azure Data Factory interface for a pipeline named 'df01121'. The left sidebar shows the 'Factory Resources' tree with 'Pipelines' expanded, and 'set variable' selected. The main canvas shows the 'set variable' activity configuration. The 'Activities' pane on the left lists various activities, with 'Set variable' selected. The 'Parameters' tab is active, showing a table of parameters: length, area, and breadth, all of type 'Int' and default value 'Value'. The 'Properties' pane on the right shows the 'General' tab with 'Name' set to 'set variable' and a description field. The 'Annotations' section is empty.

**Parameters**

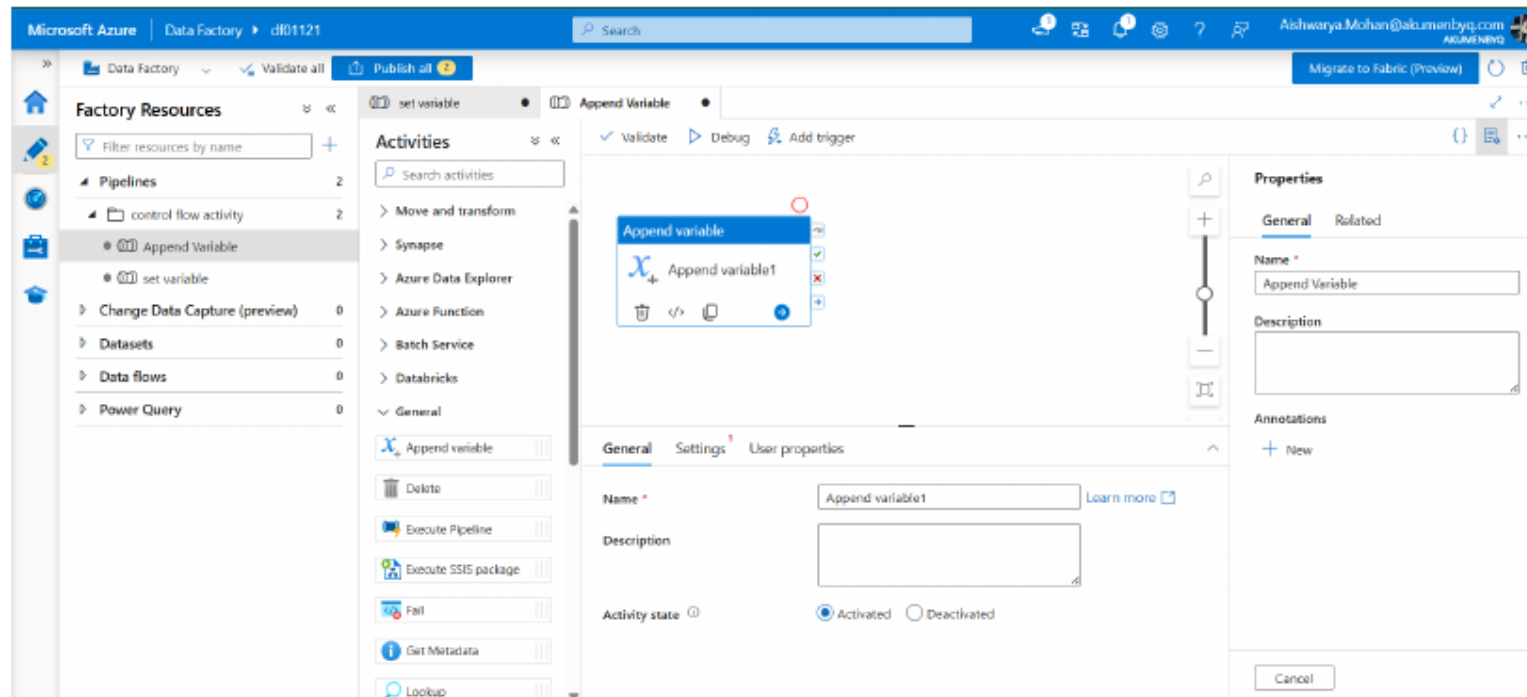
| Name    | Type | Default value |
|---------|------|---------------|
| length  | Int  | Value         |
| area    | Int  | Value         |
| breadth | Int  | Value         |

## Step 11: Append Variable Activity

Create a new pipeline (Append Variable).

Declare an Array variable (e.g. marks) with initial values.

Add Append variable activity to append a new value, then pass output to Set variable to view it.



## Step 11 (continued): Append Variable – Array Result

After appending, the array contains the original values plus the new value.  
The final array is visible in the Set variable output.

The screenshot shows the Microsoft Azure Data Factory interface. The top navigation bar includes 'Microsoft Azure', 'Data Factory', and 'dh01121'. The left sidebar shows 'Factory Resources' with a search bar and a list of activities: Pipelines (2), control flow activity (2), Append Variable (selected), set variable, Change Data Capture (preview) (0), Datasets (0), Data flows (0), and Power Query (0). The 'Activities' pane on the right lists various activities like Move and transform, Synapse, Azure Data Explorer, Azure Function, Batch Service, Databricks, and General. The 'Append Variable' activity is selected, and its configuration window is open. The 'Variables' tab is active, showing a table of variables to be appended.

| Name   | Type  | Default value    |
|--------|-------|------------------|
| marks  | Array | ["30","45","29"] |
| output | Array | ["48"]           |

The 'Properties' pane on the right shows the 'General' tab with the 'Name' field set to 'Append Variable' and a 'Description' field. The 'Annotations' section has a '+ New' button. At the bottom right, there are 'Cancel' and 'Show desktop' buttons.

## Step 11 (continued): Viewing Append Variable Output via Set Variable

Connect Append variable → Set variable.

The output shows the complete array after the append operation.

The screenshot displays the Microsoft Azure Data Factory interface for a pipeline named 'Append Variable'. The 'Activities' pane on the left shows the 'Append variable' activity connected to the 'Set variable' activity. The 'Output' pane shows the JSON output of the 'Append variable' activity, which is an array of values: ['30', '45', '29', '90']. The 'Properties' pane on the right shows the 'Name' of the variable as 'Append Variable'.

**Factory Resources**

- Pipelines: 2
- control flow activity: 2
- Append Variable
- set variable
- Change Data Capture (preview): 0
- Datasets: 0
- Data flows: 0
- Power Query: 0

**Activities**

- Append variable
- Delete
- Execute Pipeline
- Execute SSIS package
- Fail
- Get Metadata
- Lookup
- Stored procedure
- Script
- Set variable
- Validation
- Web

**Output**

```
{
  "name": "output",
  "value": [
    "30",
    "45",
    "29",
    "90"
  ]
}
```

**Properties**

**General**

Name: Append Variable

Description:

Annotations: + New

Run start: 8/18/2025, 6:26:35 PM

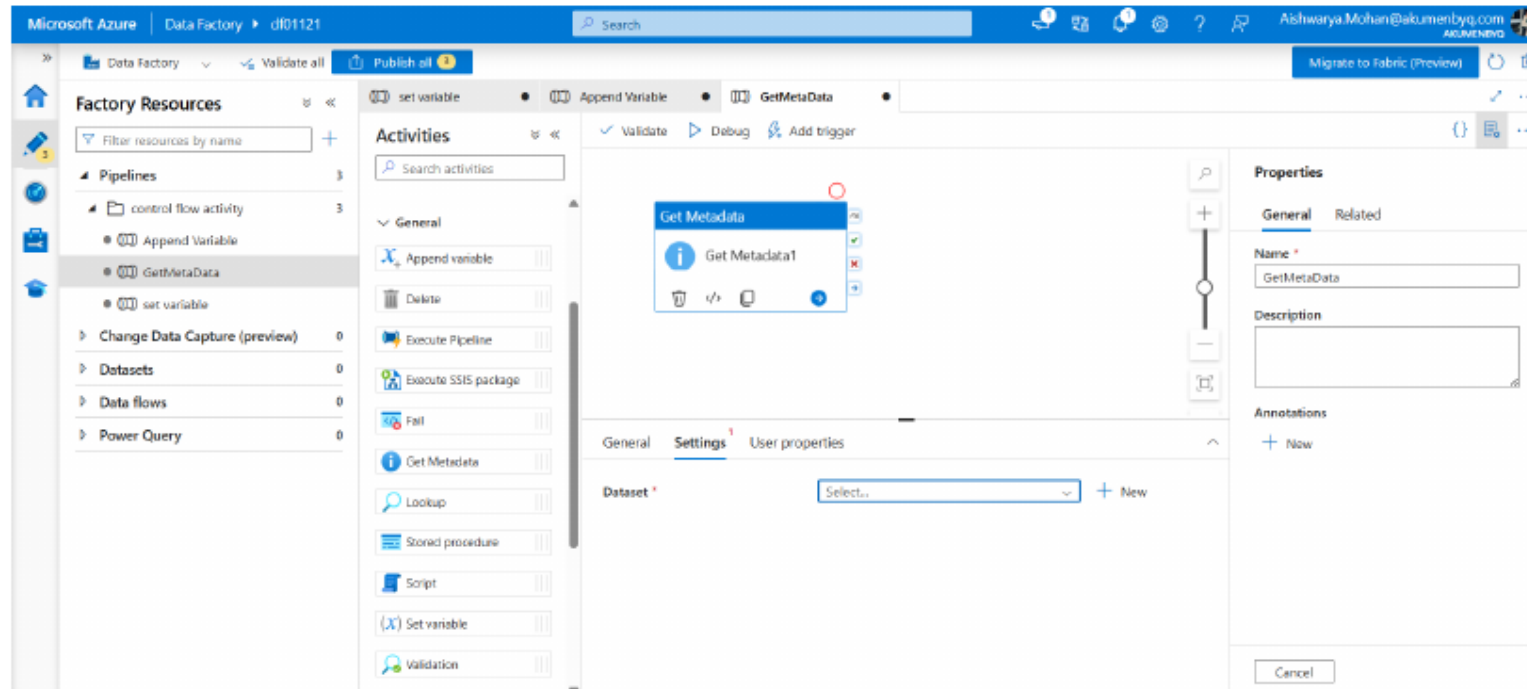
Cancel

## Step 12: Get Metadata Activity

Create a new pipeline (GetMetaData).

Add Get Metadata activity and select a dataset (e.g. DelimitedText1).

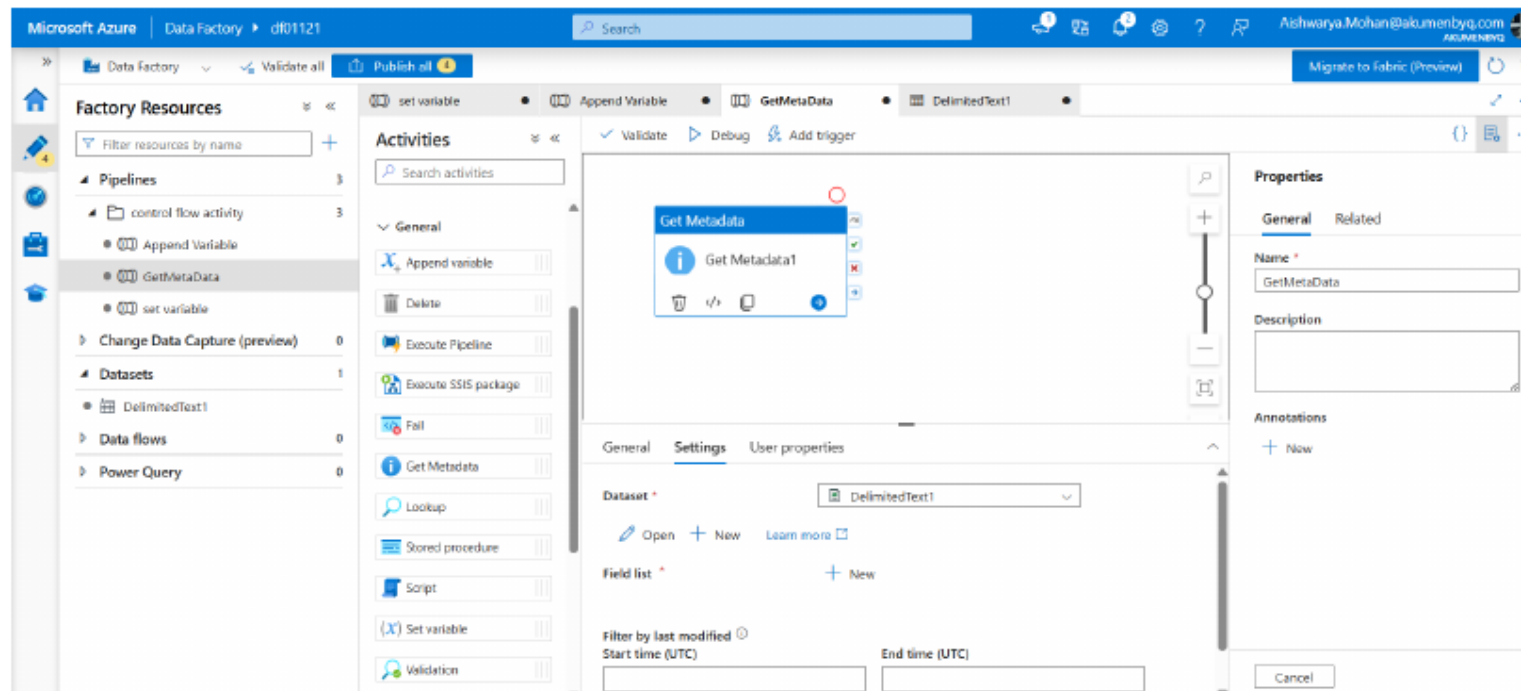
Choose the fields to retrieve. Pass the output to a Set variable activity to view it.



## Step 12 (continued): Get Metadata – Dataset and Field List

Dataset is selected and Field list is configured.

The metadata returned can be inspected via the connected Set variable activity.



## Step 13: Execute Pipeline Activity

Add Execute Pipeline activity to the parent pipeline.

Select the child pipeline to execute and pass any required parameters.

Capture the output of the child pipeline (using Set variable or Pipeline return value)

so the result is visible. Both parent and child pipelines should show Succeeded status.

The screenshot displays the Microsoft Azure Data Factory console. The top navigation bar shows 'Microsoft Azure | Data Factory | Dafa01121'. The left sidebar contains 'Factory Resources' with a search bar and a list of resources: Pipelines (2), Datasets (2), Data flows (0), and Power Query (0). The 'Pipelines' section is expanded, showing 'copy data' and 'Execute pipeline'. The 'Activities' section on the right lists various activities like Synapse, Azure Data Explorer, Azure Function, Batch Service, Databricks, and General. The 'Execute pipeline' activity is selected, and its configuration is shown in the main pane. The 'Output' tab is active, displaying the 'Pipeline status' as 'Succeeded' with a green checkmark. Below this, a table shows the execution details for 'Execute Pipeline1'.

| Activity name     | Activity st... | Activit...       | Run start              | Duration | Integration runtime |
|-------------------|----------------|------------------|------------------------|----------|---------------------|
| Execute Pipeline1 | ✓ Succeeded    | Execute Pipeline | 8/19/2026, 11:59:57 AM | 2s       |                     |

## Summary

All Control Flow activities have been successfully configured and executed:

- Copy Data activity with column mapping
  - Delete activity with logging settings
  - Manual, Schedule, Tumbling Window and Storage Event triggers
- Set Variable – area calculation (fixed values + parameterised runtime input)
  - Append Variable
  - Get Metadata
  - Execute Pipeline

This project demonstrates practical use of Azure Data Factory Control Flow activities on the Microsoft Azure platform.

Aishwarya Mohan